

Instructions to set up WSL2 and PyTorch

Last Updated: 2/17/2026

Prereqs:

- Windows 11
- NVIDIA GPU (newer 10 series)
- Latest NVIDIA windows driver (go to the NVIDIA driver website and install latest, usually no need to delete previous driver)
- Having Windows Terminal installed from the Microsoft store will make things much easier

1. Install / Update WSL2

- a. Open PowerShell Run as Administrator and run these commands

```
powershell  
  
wsl --install  
wsl --update  
wsl --set-default-version 2
```

*Note: <wsl --set-default-version 2> is not necessary, wsl2 is generally the default installation of wsl

2. Install Ubuntu Linux distro(Best for NVIDIA CUDA Toolkit)

```
powershell  
  
wsl --install -d Ubuntu
```

This might require you to reboot your computer to finish installation

3. Verify GPU is visible inside WSL

- a. In a new windows terminal window, run <wsl> to start wsl. This spins up a lightweight VM that runs a Linux Kernel on top of your windows environment.
- b. Run below code

```
bash
```

```
nvidia-smi
```

This should return your NVIDIA driver version and the NVIDIA gpu the kernel can detect. This should match with the GPU of your device. *Note: WSL uses the drivers on your Windows level to interface with your GPU, do NOT reinstall drivers in the WSL/Linux environment as that can cause issues

If this does not work, it likely means your NVIDIA drivers are not updated to the latest. Go to the web and download to your device normally.

4. Install CUDA toolkit within WSL (Pytorch comes with the needed parts from the CUDA toolkit to build without this, but this gives better functionality/makes debugging easier). In WSL env run:

```
bash
```

```
sudo apt update
```

```
sudo apt install -y build-essential
```

Then follow the instructions in the CUDA WSL guide:

<https://docs.nvidia.com/cuda/wsl-user-guide/index.html>

Short run down of above link:

- Remove old GPG key
- Install Linuxx86 CUDA Toolkit using WSL-Ubuntu Package
 - https://developer.nvidia.com/cuda-downloads?target_os=Linux&target_arch=x86_64&Distribution=WSL-Ubuntu&target_version=2.0&target_type=deb_local
 - Operating System: Linux
 - Architecture: x86_64
 - Distribution: WSL-Ubuntu
 - Version: 2.0
 - Installer Type: deb(local)

This will return a set of instructions to run in your WSL instance.

5. Install Pytorch

In Ubuntu:

```
bash

python3 -m venv ~/venvs/torch
source ~/venvs/torch/bin/activate
pip install --upgrade pip
```

IMPORTANT: below replace cu124 with cu126, that installs a more updated version.

```
bash Copy code

pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu124
```

6. That should be the end, to confirm that PyTorch sees your GPU run the below code:

```
bash Copy code

python - <<'PY'
import torch
print("torch:", torch.__version__)
print("cuda available:", torch.cuda.is_available())
print("gpu:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else None)
PY
```